

Object Recognition Based Robust Mapping for the Autonomous Mobile Robot under Dynamic Environment

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Abstract—Simultaneous Localization and Mapping (SLAM) is an essential technique for autonomous mobile robot. 2D mapping has been greatly developed recently and widely used in navigation tasks since it is easier to be generated and has fewer computation amount than 3D mapping. However, 2D information is not enough to reflect real 3D environment. This paper proposes a 2D mapping method by using immobile area grid map method. 3D objects are detected by 3D sensor and the object recognition results are reflected to the 2D map by a probability model, in which the object recognition results are used to calculate the degree of using observed information to generate the map. In this way, 3D objects are reflected to the map and potential mobile objects are deleted from the map. The effectiveness of our proposed method is proven by conducting experiments under an indoor environment, where 3D objects (e.g. desks and chairs) and human beings are all contained. With the proposed method, the generated 2D map shows the areas of 3D objects well and deletes the potential moving objects (static human beings) correctly.

Index Terms—object recognition, SLAM, dynamic environment, 2D mapping, 3D information integration

I. INTRODUCTION

In recent years, SLAM has become one of the most fascinating research fields [1], [2]. Localization and mapping are two essential information for any autonomous mobile robot working in human living environment. As the basic technique for path planning and activity planning, the generated map can also be used for robot navigation later in the same environment [3]. Laser scanners have been widely used for SLAM [4], [5] as they usually have high accuracy to detect distances in front and have a wide view angle. For example, Wang *et al.* [4] successfully realized SLAM under crowded environment and generated a map without moving objects. However, conventional methods still have a few problems to be solved. Firstly, the accuracy of localization needs to be improved. The odometry information gotten from the mobile robot can be used for localization. Whereas, its accumulative error caused by the differences between the two wheels of the robot, road condition or sensor noise will be fatal with time going on. Iterative closest point (ICP) [6] method is usually used for localization in SLAM process by matching the point cloud gotten from the laser sensor with that in the global

map. However, there may be many matched places in the map. Especially under familiar environment like the corridor, the point clouds may be matched everywhere. Secondly, the generated map based on laser scanner sensor only contains objects information from the plane where the sensor is located. It is hard to provide measurement in 3D environment where objects are in different heights. For example, desks and chairs are normal in indoor environment, but a 2D scanner sensor may only detect their legs as objects, which turn to four points on the generated map. Collision may happen during autonomous navigation for the robot. These problems can be solved by using multiple 2D sensors or 3D sensors to detect the environment [7]–[9]. Unfortunately, the accuracy of multiple 2D sensor is still limited by the number of sensors, and the cost of multiple 2D sensors or 3D sensors with wide view angle are usually high. Recently, 3D sensors with low cost like Microsoft Kinect sensor have been released but the view angle is still narrow and the information is insufficient for robot localization. Research in [10] tried to rotate a 2D scanner sensor by a motor to detect 3D information, and the real time performance is limited. Thirdly, most of conventional researches [12], [15], [20], [23] tried to remove the moving objects from the generated map, but it takes a lot of time to remove potential moving objects that are static at the moment but actually able to move in the future by these methods. Ito *et al.* [16] tried to deal with this problem by using the immobile area grid map method, in which a probability model of describing the degree of the observed object to generate the map is added. The static human beings are removed from the map quickly when they move. However, the object recognition results are limited in their method by only using laser sensors.

In this paper, an object recognition based robust mapping method for the autonomous mobile robot under dynamic environment is proposed. SLAM is realized by integrating the information of a laser scanner sensor called Laser Range Finder (LRF) and a Kinect 3D sensor. The LRF sensor is used to generate the basic 2D map to take advantage of its wide sensing range, and the Kinect sensor is used to recognize different kinds of 3D objects and reflect them to the basic

2D map. In this way, the generated 2D map contains 3D information so that the environment can be better reflected to the 2D map. The idea of sensor placement is familiar with Lin *et al.* [11], but different from their work, the generated map is more robust in removing potential moving objects. The immobile area grid map based SLAM method is used to represent the map by an immobile probability for each grid. The map update parameter is set adaptively from the certainty of identification result. Potential moving objects are not mapped if the identification results are corresponding to moving objects. The experimental results show that the proposed method is more robust to generate a 2D method containing 3D information while deleting potential mobile objects from the map.

This paper is organized as follows. Section I shows the introduction of the paper. Section II explains the method of improving the accuracy of robot self-localization. In section III, the idea of integrating 2D and 3D information is proposed. Section IV explains the probabilistic model of mapping under dynamic environment, by using immobile Area Grid Map Based SLAM. The experimental results and analyses are shown in section V. Finally, section VI presents the conclusions and future work.

II. ROBOT SELF-LOCALIZATION

To guide the users correctly, the robot needs to locate itself accurately. The ICP method can be used for localization [6]. Its working process is shown in Fig. 1. The newly detected distance data by the laser sensor is matched with the model data, as shown in Fig. 1 (a). These two types of data should be similar throughout continuous frames, and the location with the closest match can be determined by minimizing the distance between corresponding points, as shown in Fig. 1 (b). The calculated positional relation shows the motion of the robot, and the localization of the robot can also be calculated as shown in Fig. 1 (c). However, this method has the problem that there may be multiple matched places with the map. For example, in the corridor, the model data are similar with the new detected sensor and matched with the detected distance data everywhere, as shown in Fig. 2. The odometry information received from the mobile robot can also be used for localization. Notably, the accumulative error caused by the differences between the two wheels of the robot, road condition, or sensor noise will be fatal as time goes on. Thus, these two types of information are integrated by using the Monte Carlo Markov Chain (MCMC) [17] particle filter method. During the prediction step, the motion information obtained from odometry is used to predict the position of the robot, mixed with Gaussian noise in order to use it as the motion model. During the evaluation step, the prediction results were evaluated and re-weighted according to the matching rates between the detected distance data and the model data, same with the ICP method. The robot position then can be calculated as the re-weighted mean of all prediction results.

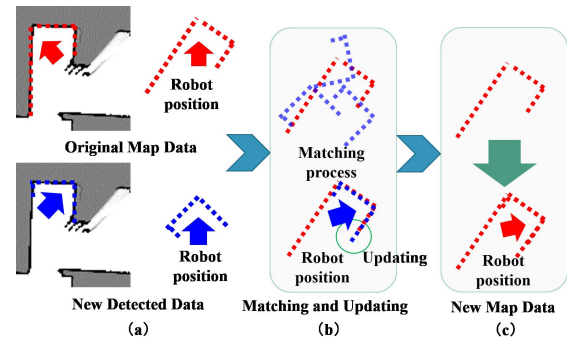


Fig. 1. Working process of the iterative closest point (ICP) method.

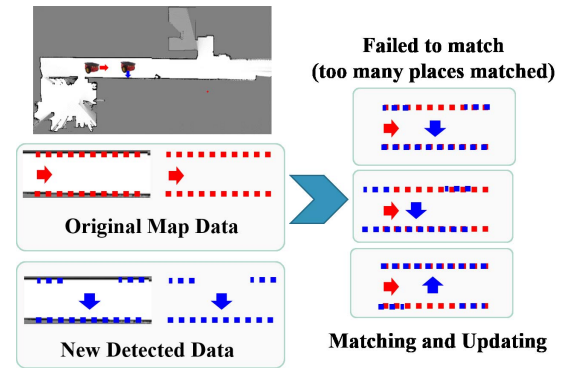


Fig. 2. Matching in multiple places when using the iterative closest point (ICP) method.

III. INTEGRATION OF 2D AND 3D INFORMATION

To understand the environment around the robot, distance sensors are usually used for SLAM. In the proposed system, the LRF sensor is used for getting the 2D distance information, and Kinect sensor is used for getting the 3D information around the robot. LRF sensor is chosen as the UTM-30LX made by Hokuyo Co.. It can sense the range of 270° in front of the robot with the maximum distance of 10m. The degree step is 0.25°. The LRF sensor is shown in Fig. 3 (a) and its sensing range is shown in Fig. 3 (b). Kinect sensor can get the color and depth information at the same time in front of the sensor. It is made by Microsoft Co. and the Kinect V2 is chosen for the experiment. The sensing range of the Kinect sensor is 70° in horizontal and 60° in vertical with the maximum distance of 8m. The color image resolution of the Kinect sensor is 1920*1080 pixels. The Kinect sensor is shown in Fig. 4 (a) and its sensing range is shown in Fig. 4 (b).

LRF has a large sensing range but it can only detect the distance information in a horizontal plane. The objects in front of the robot can be detected well on the sensing plane. For the scene shown in Fig. 5 (a), the object detection result is shown in Fig. 5 (b). Kinect can detect 3D distance information and

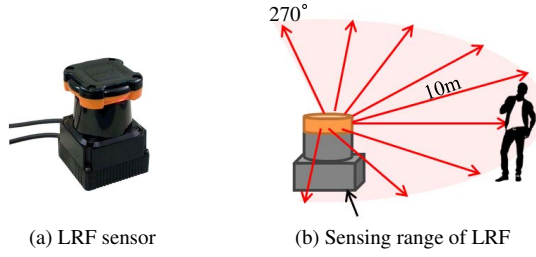


Fig. 3. Introduction of LRF sensor.

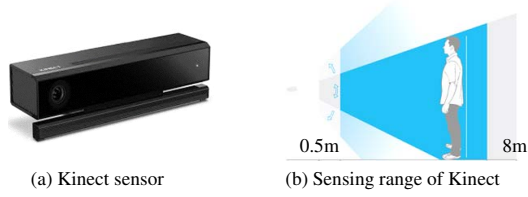


Fig. 4. Introduction of Kinect sensor.

color information at the same time, but its sensing range is limited. Thus, these two kinds of information are integrated for generating a robust map, which can reflect the objects well in the space. The distance information of Kinect sensor (3D information) is projected to the ground plane to integrate with that gotten from LRF sensor (2D information). Meanwhile, the human detection can be easily processed by Kinect sensor [18]. From the distance information of Kinect sensor, the planes can be detected by using Point Cloud Library (PCL) [19]. After the planes are detected, the ceiling plane can be separated as the highest horizontal plane over 2.5m, and the wall planes can be recognized as the vertical planes that connected to the ceiling plane. For the scene shown in Fig. 6 (a), the human detection result is shown in Fig. 6 (b). After combining these two kinds of information on the 2D map, the result is shown in Fig. 7. The human beings, walls and 3D obstacles like desks are well reflected on the map.

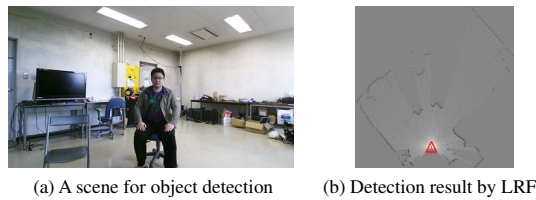


Fig. 5. Environment sensing result by LRF.

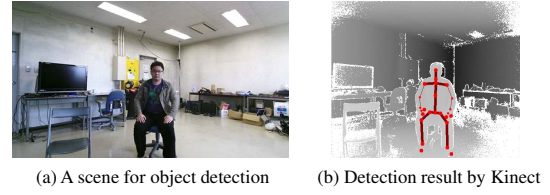


Fig. 6. Environment sensing result by Kinect.

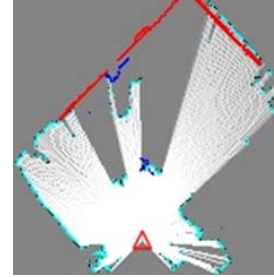


Fig. 7. Environment sensing result after integrating two kinds of information.

IV. IMMOBILE AREA GRID MAP BASED SLAM

The occupancy grid map used in SLAM is a conventional method which makes a map by an occupancy probability in each grid. When the occupancy probability is high, it means that the probability of existing an object for the grid is high. When the occupancy probability is low, it means that the probability of existing an object for the grid is low, and the probability of noting for the grid is high. The robot keeps sensing the environment around and updating the occupancy probability in each grid. In this way, the areas where objects exist or not can be expressed by probability. However, this method has the problem that even the moving objects would be registered on the map if they are observed in a high probability. As the robust environment map should not contain moving objects and potential moving objects (like a human being keeps still), other methods which can delete the moving objects are used for the researches [20]- [23]. To solve the problem, the immobile grid map [16] is applied to generate the map. The environment is divided into grids first, same with the occupancy grid map method. However, the probability for each grid is calculated by the probability of suitable immobile area instead of the occupancy probability. When the occupancy grid map is used, the updating weigh will increase in the exponential way if the object existing probability is high. Moreover, the observed objects might be moving objects under the dynamic environment. The map will be generated correctly only when the objects are still objects as the moving objects should not be shown in the environment map. For the immobile area grid map method, the map updating weight can be adjusted adaptively even if the object existing probability is high, by using a changeable parameter that controlled by

the object recognition results.

The event that one grid is immobile area is set as I , and the event that some objects are observed on the same grid is set as O . The immobile area probability $P_t(I)$ can be expressed as Equation (1) at time t .

$$P_t(I) \propto P_t(O) \cdot P_{t-1}(I) \quad (1)$$

Here, $P_t(O)$ is the object existing probability. From Equation (1), the probability $P_t(I)$ that one grid is immobile area at time t can be calculated from the observed object existing probability $P_t(O)$ at time t and the immobile area probability P_{t-1} for the same grid at time $t - 1$. The event I can exist depending on the event O can be observed.

Meanwhile, the event that one grid is not immobile area is set as $\bar{I}$, and the event that any objects cannot be observed on the same grid is set as $\bar{O}$. The non-immobile-area probability $P_t(\bar{I})$ can be expressed as Equation (2) at time t .

$$P_t(\bar{I}) \propto P_t(\bar{O}) \cdot P_{t-1}(\bar{I}) + P_t(O) \cdot P_{t-1}(\bar{I}) \quad (2)$$

From Equation (2), the event $\bar{I}$ can exist depending on the event O or the event $\bar{O}$ can be observed. It means that the sum of the cases that the event $\bar{O}$ that nothing is observed happens and the event O that some objects are observed as moving objects happens. From Equation (2), the following equation can be gotten.

$$P_t(\bar{I}) \propto P_{t-1}(\bar{I}) \quad (3)$$

From Equation (3), the event $\bar{I}$ exists only depending on a constant value. To divide the Equation (1) by Equation (3), the Equation (4) can be gotten.

$$\frac{P_t(I)}{P_t(\bar{I})} \propto \lambda \cdot P_t(O) \cdot \frac{P_{t-1}(I)}{P_{t-1}(\bar{I})} \quad (4)$$

Here, λ is the constant coefficient. The updating coefficient for immobile area probability is defined as β , Equation (5) can be gotten.

$$\beta = \lambda \cdot P_t(O) \quad (5)$$

Here, λ is defined as the variable parameter that is controlled by the object recognition results. The observed objects are recognized as the immobile areas or moving objects first, and the results are used for setting the value of λ . Then we can adjust the value of β by the value of λ . The potential moving objects can be deleted for updating the map by adjusting the value of λ .

The robot senses the environment by its sensors, and the observed objects are recognized first. The value of λ is adjusted depending on the recognition results. If the results are unknown objects, the object is ambiguous and $P_t(O) = 0.5$. The immobile area probability should not be changed as the observed information is meaningless for updating the map. Thus, the value of λ is set as 2 to make sure the updating coefficient for immobile area probability $\beta = 1$. If the recognition results are moving objects, the value of λ is set smaller than 2 so that the observed objects are not used for updating the map. If the recognition results are still objects, the value of λ is set bigger than 2 so that the observed objects

are used for updating the map by a big updating coefficient and the objects can be quickly reflected on the map.

The value of λ is set based on the likelihood of the object recognition results. The likelihood of the object recognition results expresses how much the recognition result can be trusted. When the likelihood is low, the objects are better to be thought as unknown objects, and the value of λ is around 2. With the increasing of the likelihood, the value of λ increases from 2 if the objects are recognized as still objects, and the value of λ decreases from 2 if the objects are recognized as moving objects.

V. GLOBAL MAPPING OF THE WORK ENVIRONMENT

The map of the work environment is generated in advance. The mobile platform is chosen as the PIONEER 3-DX made by MobileRobots Co.. The height of the LRF sensor is set as 32cm and the Kinect sensor is set as 100cm above the ground on the robot. The Laser Range Finder (LRF) sensor is used for getting the 2D distance information and the Kinect sensor is used for getting the 3D distance information.

Firstly, the experiment is conducted in a room where kinds of objects and a person are inside. The person moved from still status during the experiment. The scene is shown in Fig. 8. The generated map by the proposed method and conventional method [11] are shown in Fig. 9 and Fig. 10 separately. The conventional SLAM method is based on occupancy grid map, which is most widely used by researchers since the open source is released already and possible to apply for kinds of indoor environments. In Fig. 9 and Fig. 10, the black points show the immobile areas, the white points show the mobile areas, and the gray points show the unknown areas in the map. By comparing the two results, the chairs and desks are detected as some areas in the proposed method, but only detected as some points in the conventional method. These results make sure that the robot cannot go through in these areas so that collisions can be avoided. By our proposed method, the person can be detected well from the beginning even if he was static, and these areas in the map are not updated as objects. On the other hand, the conventional method can only recognize the person after he started to move. Moreover, the chair that the person sat before moving can be also reflected to the map well by the proposed method.

The proposed method is used for generating the map of the lab environment where the guide robot will work. The mapping result is shown in Fig. 11. We observe that the humans are all deleted from the map, no matter they are static or moving. The areas of desks and chairs are shown as areas with objects, rather than some points (e.g. only the legs of desks or chairs). Moreover, the mobile objects are deleted from the map in a fast speed. This map is good enough for path planning of mobile robots.

VI. CONCLUSION

This paper proposed a robust 2D mapping method containing 3D object recognition results. 3D information are reflected to the basic 2D map to make the 2D map more accurate and

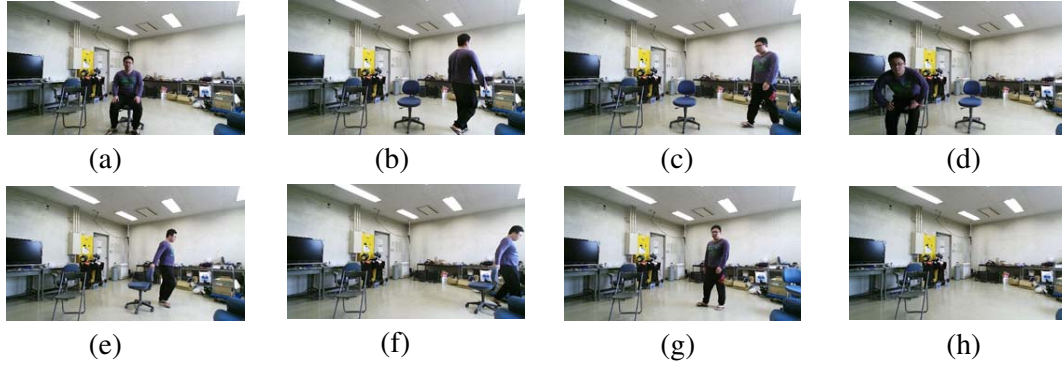


Fig. 8. The scene of the environment.

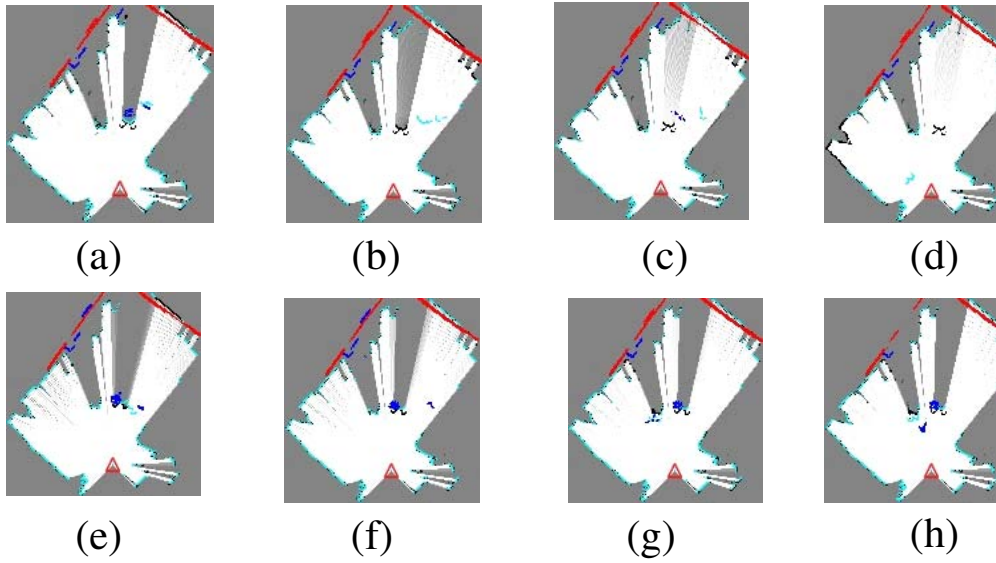


Fig. 9. Mapping results by the proposed method.

robust. For example, desks and chairs are reflected as four points on the map by using laser sensor, but the planes of the desks and chairs are recognized by the 3D sensor and reflected as areas in the 2D map. This helps the robot to improve safely when moving around. The map is generated and updated by using immobile area grid map. The 3D object recognition results are used to calculate the degree of using the observed objects to update the map by probability. In this way, the potential mobile objects are quickly deleted from the map. The effectiveness of the proposed method is proven by experiments. This method has great performance under indoor dynamic environment containing multiple objects (e.g. desk and chair) and potential mobile objects (e.g. humans who keep static in the beginning).

Future work will focus on recognizing more objects and improving the accuracy of object recognition result. The attributes of the objects will also be reflected to the map to make it more easy to use for path planning of autonomous mobile robots. For example, the robot should not go through the space under a desk, and should not stand by behind a door. In this way, the robots can move more naturally like human beings and the social acceptability of robot will be improved.

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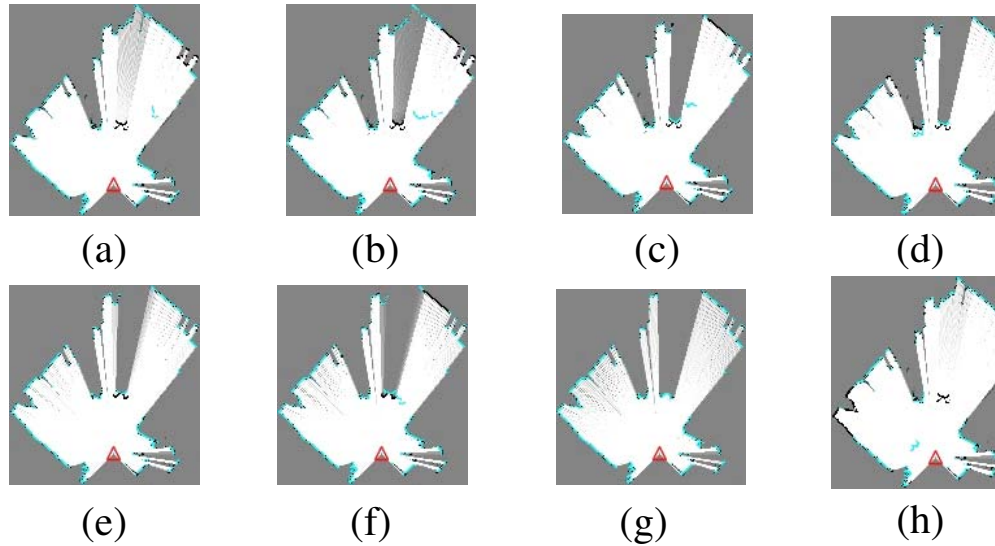


Fig. 10. Mapping results by the conventional method.

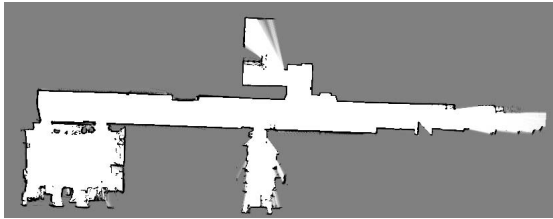


Fig. 11. Mapping results of the lab environment.

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